

Introduction to $\text{\LaTeX}$

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Agenda

Introduction to
 $\text{\LaTeX}$
Georgios
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General

Text

Greek

Math

Publications

Topics

- General
- Text
- Greek
- Math
- Publications



Agenda

Introduction to
 $\text{\LaTeX}$
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Topics

- General
- Text
- Greek
- Math
- Publications



Who is Tyler Durden?

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Drakopoulos

General

Text

Greek

Math

Publications

What is $\text{\LaTeX}$?

- Typesetting (eco)system
- Supports a markup language
- Concentrates on content and structure
- Configurable style
- Based on $\text{\TeX}$ typesetting engine

What is not $\text{\LaTeX}$?

- Text editor
- Graphical interface

Form vs content

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General

Text

Greek

Math

Publications

Form

- Font type
- Boldface or italics
- Bullet type

Content

- Text
- Equations
- References
- Sections

Advantages - software

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L^AT_EX

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General

Text

Greek

Math

Publications

Open source

- Free
- Active user community
- Huge package collection

Inherent interoperability

- Runs in every known operating system

Interfaces with production software

- Doxygen
- matlab
- texinfo

Advantages - functionality

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L^AT_EX

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Drakopoulos

General

Text

Greek

Math

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L^AT_EX source code is (hyper)text

- MIME: application/x-tex
- Easy text recovery
- Use your favorite text processor
- Use special environments for L^AT_EX
- Use version control systems
- Use Beamer for slides

Common in academia

- The de facto academic standard

Advantages - typesetting

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General

Text

Greek

Math

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Large texts

- No Cupertino syndrome
- Table of contents
- Table of figures
- Easy figure handling
- Easy text structure
- Intuitive section, table, and figure numbering
- Automatic references
- Source code across multiple files
- Bibliography
- Macros and new commands
- Templates

Disadvantages

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Drakopoulos

General

Text

Greek

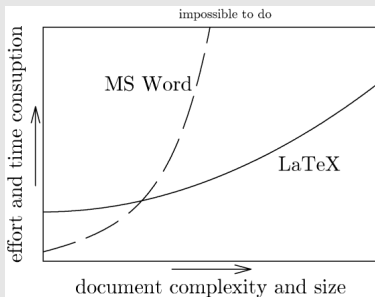
Math

Publications

Learning a programming language

- Compilation
- Debugging
- Tools and environments

Take home message



Environments

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General

Text

Greek

Math

Publications

L^AT_EX-specific

- TeX studio
- TeXworks
- TeXnic center
- LyX

Generic

- Linux editors (vim, emacs, . . .)
- Windows



Typesetting engines

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General

Text

Greek

Math

Publications

pdfT_EX and pdfL^AT_EX

- Usually the starting point

X_YT_EX and X_YL^AT_EX

- Inherent Unicode support
- Particulary popular in mac

LuaT_EX and LuaL^AT_EX

- Inherent Unicode support
- Use Lua code directly
- Needs work for multilingual support

Agenda

Topics

- General
- **Text**
- Greek
- Math
- Publications



Article

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Drakopoulos

General

Text

Greek

Math

Publications

$\text{\LaTeX}$ code

```
\documentclass[10pt]{article}
% preamble
\author{Ludwig van Beethoven}
\title{\TeX und \LaTeX}
\date{\today}

%text body
\begin{document}
\maketitle

\paragraph{} Hello world!

\end{document}
```

Report

Introduction to
L^AT_EX
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Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\documentclass[10pt]{report}
\author{Wolfgang Amadeus Mozart}
\title{A \LaTeX{} report}
\date{\today}

\begin{document}
\maketitle

\chapter{Number one}
\paragraph{} This is first chapter.

\chapter{Number two}
\paragraph{} This is second chapter.
\end{document}
```

Document class

Every L^AT_EX document starts with

- `\documentclass [<options>] {<class>}`
- `[<optional>]: optional argument`
- `{<mandatory>}: mandatory argument`
- `<class>: class name`

Common L^AT_EX classes

- article
- report
- book

Environments

Basic building block

- Anything between `\begin{<name>}` and `\end{<name>}`

Properly nested

```
\begin{<name1>
  \begin{<name2>}
    % arbitrarily deep nesting
  \end{<name2>}
\end{<name1>}
```

Improperly nested

```
\begin{<name1>}
  \begin{<name2>}
    % compilation error
  \end{<name1>}
\end{<name2>}
```

Bullets

L^AT_EX code

```
\begin{itemize}  
  \item one  
  \item two  
  \item three  
\end{itemize}
```

Result

- one
- two
- three

Numbered lists

L^AT_EX code

```
\begin{enumerate}  
  \item one  
  \item two  
  \item three  
\end{enumerate}
```

Result

- 1 one
- 2 two
- 3 three

Tables

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L^AT_EX
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\begin{table}[h]
  \begin{tabular}{|c|c|}
    \hline
    one      & two      \\
    three    & four     \\
    \hline
  \end{tabular}
\end{table}
```

Result

one	two
three	four

Figures

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Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Preamble

```
\usepackage[<dvips>]{graphicx}  
\usepackage{color} %optional but common
```

Text

```
\begin{figure}[h]  
\includegraphics[scale=xx]{<name>}  
\end{figure}
```

Lists

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General

Text

Greek

Math

Publications

Contents

- `\tableofcontents`

Figures

- `\listoffigures`

Tables

- `\listoftables`

Notes

- Automatically generated and updated

Source code - The old way

L^AT_EX code

```
\begin{verbatim}

    function [x,y] = my_func(a,b)
    %MY_FUNC is a matlab function
    x = min(a,b);
    y = max(a,b);
    return

\end{verbatim}
```

Notes

- Rarely used today
- Use package listings

Source code - The new way

Introduction to
L^AT_EX
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Preamble

```
\usepackage{listings}
```

L^AT_EX code

```
Source code in \LaTeX:  
\lstset{language=matlab}  
\begin{lstlisting}  
for i = 1:n  
    v[i] = i;  
end  
\end{lstlisting}
```

Result

Source code in L^AT_EX:

```
for i = 1:n  
    v(i) = i;
```

thebibliography - commands

L^AT_EX code

- In text `\cite{<name>}`
- `\bibliographysyle{<style>}`
- `\begin{thebibliography}{<width>}`

Common styles

- | | |
|----------|-----------|
| • plain | • apalike |
| • abbrev | • siam |
| • alpha | • acm |

Note

- Two compilations are necessary

thebibliography - use

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L^AT_EX

Georgios

Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\documentclass[10pt]{article}
...
\begin{document}
\bibliographystyle{abbrv}
\begin{thebibliography}{1}

\bibitem{key1} Contributor, My article,
    This conference.

\bibitem{key2} Author, Another article,
    That journal.

\end{thebibliography}
\end{document}
```

bibtex - use

L^AT_EX code

```
\documentclass[10pt]{book}  
\begin{document}  
Regular \LaTeX{} text \cite{book:1900}  
\bibliographystyle{abbrv}  
\bibliography{bibfile}  
\end{document}
```

Common options

- @book
- @booklet
- @report
- @inproceedings
- @article
- @misc

bibtex - bib files

Entry

```
@book{book:1990,  
  author = {One, Author and Two, Contributor},  
  title = {A great paper},  
  publisher = {This} ,  
  year = {2013}  
}
```

Fields

- | | |
|-------------|------------|
| • author | • volume |
| • title | • magazine |
| • year | • pages |
| • publisher | • doi |

Note

- Use a doi wherever possible

References

L^AT_EX code

- Insert with `\label{<name>}`
- Use with `\ref{<name>}`
- Two compilations may be required

What can be referenced?

- Tables
- Figures
- Equations
- Chapters
- (Sub)sections

Agenda

Introduction to
 $\text{\LaTeX}$
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Topics

- General
- Text
- **Greek**
- Math
- Publications



babel

Preamble

```
\usepackage[english, greek]{babel}  
\usepackage[utf8x]{inputenc}  
\newcommand{\en}{\selectlanguage{english}}  
\newcommand{\gr}{\selectlanguage{greek}}
```

Notes

- Use `\en` and `\gr` to switch between languages
- Last language in `babel` declaration is the default



Greek letters

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Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Control sequences

<code>\alpha</code>	α	<code>\beta</code>	β	<code>\gamma</code>	γ
<code>\delta</code>	δ	<code>\epsilon</code>	ϵ	<code>\varepsilon</code>	ε
<code>\zeta</code>	ζ	<code>\eta</code>	η	<code>\theta</code>	θ
<code>\vartheta</code>	ϑ	<code>\iota</code>	ι	<code>\kappa</code>	κ
<code>\lambda</code>	λ	<code>\mu</code>	μ	<code>\nu</code>	ν
<code>\xi</code>	ξ	<code>\pi</code>	π	<code>\varpi</code>	ϖ
<code>\rho</code>	ρ	<code>\varrho</code>	ϱ	<code>\sigma</code>	σ
<code>\varsigma</code>	ς	<code>\tau</code>	τ	<code>\upsilon</code>	υ
<code>\phi</code>	ϕ	<code>\varphi</code>	φ	<code>\chi</code>	χ
<code>\psi</code>	ψ	<code>\omega</code>	ω	<code>\Gamma</code>	Γ
<code>\Delta</code>	Δ	<code>\Theta</code>	Θ	<code>\Lambda</code>	Λ
<code>\Xi</code>	Ξ	<code>\Pi</code>	Π	<code>\Sigma</code>	Σ
<code>\Upsilon</code>	Υ	<code>\Phi</code>	Φ	<code>\Psi</code>	Ψ
<code>\Omega</code>	Ω				

Agenda

Introduction to
 $\text{\LaTeX}$
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Topics

- General
- Text
- Greek
- **Math**
- Publications



How to write math?

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General

Text

Greek

Math

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Standard $\text{\LaTeX}$ modes

- $\$$ for embedded mode
- $\$$ $\$$ for display mode
- `eqnarray` and `eqnarray*` for multiple lines

amsmath

- $\$$ still in use
- `\(` and `\)` for display mode
- `align` and `align*` for multiple lines

Embedded mode

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Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

Fermat's last theorem: $x^n + y^n = z^n$ when x, y, z are integers and $n > 2$.

Result

Fermat's last theorem: $x^n + y^n = z^n$ when x, y, z , and n are integers if and only if $n = 2$.

Multiple lines

L^AT_EX code

```
\begin{align*}  
    \alpha_1 + \beta_1 &= \gamma_1 \quad \cr  
    \alpha_2 + \beta_2 &= \gamma_2  
\end{align*}
```

Result

$$\alpha_1 + \beta_1 = \gamma_1$$

$$\alpha_2 + \beta_2 = \gamma_2$$

Matrices

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L^AT_EX
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\[
A \coloneqq \begin{bmatrix}
a_{1,1} & \ldots & a_{1,n} \\
a_{2,1} & \ldots & a_{2,n} \\
\vdots & \ddots & \vdots \\
a_{m,1} & \ldots & a_{m,n}
\end{bmatrix}
\]
```

Result

$$A = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ a_{2,1} & \dots & a_{2,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix}$$

Functions with branches

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L^AT_EX
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\[  
x[n] \: = \:  
\begin{cases}  
    r^{-n}, & n \geq 0 \\cr  
    0,      & n < 0  
\end{cases}  
\]
```

Result

$$x[n] = \begin{cases} r^{-n}, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

L^AT_EX and mathematics

Introduction to
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Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Symbols

<code>\infty</code>	∞	<code>\pm</code>	$\pm$	<code>\mp</code>	$\mp$	<code>\cap</code>	$\cap$
<code>\propto</code>	$\propto$	<code>\bowtie</code>	$\bowtie$	<code>\star</code>	$\star$	<code>\cup</code>	$\cup$
<code>\geq</code>	$\geq$	<code>\leq</code>	$\leq$	<code>\ll</code>	$\ll$	<code>\gg</code>	$\gg$
<code>\oplus</code>	$\oplus$	<code>\odot</code>	$\odot$	<code>\otimes</code>	$\otimes$	<code>\times</code>	$\times$
<code>\sum</code>	$\sum$	<code>\prod</code>	$\prod$	<code>\int</code>	$\int$	<code>\oint</code>	$\oint$

Functions

<code>\sin</code>	sin	<code>\cos</code>	cos	<code>\tan</code>	tan
<code>\arcsin</code>	arcsin	<code>\arccos</code>	arccos	<code>\arctan</code>	arctan
<code>\frac{x}{y}</code>	$\frac{x}{y}$	<code>x^n</code>	x^n	<code>x_1</code>	x_1
<code>\to</code>	$\rightarrow$	<code>\lim</code>	lim	<code>\ x\ </code>	$\ x\ $

Agenda

Introduction to
 $\text{\LaTeX}$
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Topics

- General
- Text
- Greek
- Math
- **Publications**



Research paper templates

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General

Text

Greek

Math

Publications

Scientific societies

- IEEE: IEEEtran
- ACM: sigproc-sp, acm_proc_article-sp
- SIAM: siamltex

Publishing houses

- Springer: svmult, springer3
- Elsevier: elsarticle
- Wiley: WileySix, WileySev
- MITPress: mitpx

Manuals and distributions

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Georgios
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General

Text

Greek

Math

Publications

Manuals

- $\text{\LaTeX}$ reference card (GNU)
- $\text{\TeX}$ primer (TUG)
- The not so short introduction to $\text{\LaTeX}$ (Lamport/Oetiker)
- The $\text{\TeX}$ book (Knuth)

Distributions

- www.latex-project.org
- www.tug.org/texlive
- www.miktex.org
- www.ctan.org

Template use

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General

Text

Greek

Math

Publications

L^AT_EX code

```
\documentclass[10pt, conference]{IEEEtran}
```

Options

- conference: IEEE conferences
- compsocconf: computer society conferences

Notes

- Not a standard L^AT_EX class
- Download IEEEtran.cls

Title

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General

Text

Greek

Math

Publications

L^AT_EX code

```
\title{<title>}
```

Notes

- Be careful with capitalization
- Use some keywords in title
- Be concise
- No `\date` command

Authors and affiliations

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Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\author{%  
    \IEEEauthorblockN{%  
        <name>  
    }  
    \IEEEauthorblockA{%  
        <affiliation>  
    }  
}
```

Notes

- Use `IEEEauthorrefmark` for multiple authors

Keywords

Introduction to
L^AT_EX
Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\begin{IEEEkeywords}  
Keyword1; Keyword2; ...  
\end{IEEEkeywords}
```

Notes

- Be concise
- Avoid many generic terms
- Use 5-7 keywords

Abstract

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Georgios
Drakopoulos

General

Text

Greek

Math

Publications

L^AT_EX code

```
\begin{abstract}  
<text>  
\end{abstract}
```

Notes

- Be concise
- State problem
- Sketch methodology
- Explain results
- A quarter of a column

Communities

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Drakopoulos

General

Text

Greek

Math

Publications

International

- www.tug.org
- tex.stackexchange.com
- www.sharelatex.com
- wikibooks.org/wiki/LaTeX

Greek

- www.greekfont society.gr
- www.eutyp on.com

Deja vu

Introduction to
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Georgios
Drakopoulos

General

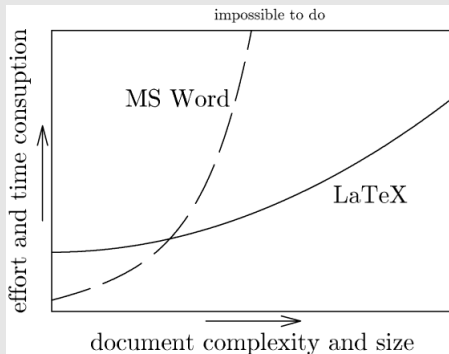
Text

Greek

Math

Publications

Take home message



Thank you for your attention

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$\text{\LaTeX}$

Georgios
Drakopoulos

General

Text

Greek

Math

Publications

Questions?

